

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	S.Mythili	Reg. No.:	1924 11250
Section / Batch:		Date:	16/07/2026

Instructions to Students

Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
Clearly show all steps in derivations and complexity analysis.
Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
Justify each step in recurrence solving using appropriate methods.
Clarity of presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Multi-Stage Authentication System

A security authentication algorithm reduces the problem size by half at each recursive stage and performs constant verification work at every level. Apply the substitution method to solve the recurrence $T(n) = T(n/2) + 1$. Show the detailed recursive expansion and derive the final asymptotic complexity. Analyze why the algorithm remains efficient over multiple levels of recursion. Interpret the result using asymptotic notation.

Q2. Autonomous Vehicle Navigation System

An autonomous navigation system processes route data using the recurrence $T(n) = 2T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how this affects performance as map data increases. Interpret efficiency using asymptotic notation.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						

	Overall Total (100)	
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Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1 Multi-Stage Authentication System

Problem Statement

A security authentication algorithm reduces the problem size by half at each recursive stage and performs constant verification work at every level.

Given recurrence:

Use the Substitution Method to solve the recurrence and determine its asymptotic complexity.

Understand the Recurrence

The problem size is reduced from n to $n/2$ at every recursive call.

Each recursive level performs constant work ($\uparrow$) for verification.

The recursion continues until the input size becomes 1

Given recurrence:

Step 1 Recursive Expansion (Substitution Method)

Expand the recurrence repeatedly.

Substitute :

Substitute again:

After k expansions:

Step 2: Find the Stopping Condition

Recursion stops when:

Taking log base 2:

Step 3: Substitute Back

Since (constant),

Ignoring constants,
Efficiency Analysis

At every recursive level, the problem size becomes half.

Only constant work ($\uparrow$) is done at each level.

Number of recursion levels = .

Therefore, total work grows very slowly even for large inputs.

Asymptotic Interpretation

Interpretation:

The authentication algorithm has logarithmic time complexity, meaning the execution time grows slowly as the input size increases. This ensures fast authentication even for very large datasets.

Q2. Autonomous Vehicle Navigation System

Problem Statement

An autonomous vehicle navigation system processes route data using the recurrence

Apply the Master Theorem to determine the time complexity.

Given recurrence:

Step 1: Identify Master Theorem Parameters
Compare with

Here,

Step 2: Compute

Thus,

Compare:

So,

where .

Also,

Step 3: Select the Master Theorem Case

Since

and the regularity condition is satisfied,

Step 4: Final Complexity

Performance Analysis

The navigation system divides the map into two halves at each recursive step.

However, processing each level requires work, which dominates the recursive cost.

As the map size increases, execution time grows quadratically.

Asymptotic Complexity

Interpretation:

The algorithm's running time increases proportionally to the square of the input size. For large maps, the quadratic processing cost dominates, making the algorithm less scalable than logarithmic or linear-time solutions.